ORLANDO HEALTH
1414 Kuhl Ave.
Orlando, FL 32806

DEPARTMENT GUIDELINE

Guideline Title: **Pediatric Allergic Reaction/Anaphylaxis**	Guideline Number: **GUID-3203-PED002**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** Recognition of a pediatric patient presenting with anaphylaxis; provide and maintain adequate airway control, oxygenation, ventilation, and end-organ tissue perfusion; and to provide emergent care and expeditious transport to the nearest appropriate medical facility.

II. **DEFINITIONS:**
   a. Mild reaction: itching/hives
   b. Moderate reaction: dyspnea, wheezing, chest tightness, facial swelling
   c. Severe Reaction: < 90 mmHg, MAP < 65 mmHg, stridor, severe respiratory distress, vomiting/diarrhea

III. **DEPARTMENT GUIDELINE:**
   a. Establish care as defined by *GUID3203-G002 – General Patient Care* & *GUID3203-PED001 – Pediatric General Management.*
      i. Assess and manage airway, breathing, and circulation.
      ii. Provide continuous EKG and oximetry monitoring, frequent vital signs.
   b. While assessing the patient, attempt to determine the etiology of the allergic reaction.
      i. If medication infusion reaction suspected, stop the infusion.
   c. If the patient is experiencing a severe life-threatening reaction, or shows signs of shock consider
      i. **Epinephrine (1:1,000) 0.01 mg/kg IM q15 min** (0.01ml/kg) **-** Max single dose 0.3 mg every 3-5 minutes or sooner. Anterior thigh is preferred.
         1. Massage injection site vigorously for 30-60 seconds
   d. Obtain vascular IV/IO access and optimize volume resuscitation per *GUID3203-PED010 - Pediatric Hypotension.*
      If the patient is hypotensive or displays signs of inadequate tissue perfusion, administer a fluid bolus of 20ml/kg 0.9% NS. Repeat as necessary.
      i. If the patient remains hypotensive after multiple fluid boluses, consider initiating **Epinephrine (4 mg/250 ml D5W) infusion at 0.05 mcg/kg/min** and titrate every 2-3 min by 0.05mcg/kg/min. Max rate 2 mcg/kg/min.
         1. If patient is on beta blockers and appears resistant to epinephrine treatment, consider **Glucagon 20 mcg/kg IV** slow over 5 minutes. Observe for vomiting.

e. Perform a rapid assessment of the patient's airway
    i. Intubation should be performed if stridor or respiratory distress due to upper airway compromise, significant edema of tongue, oropharyngeal tissues, or voice alteration per *GUID3203-G008 - Rapid Sequence Induction for Endotracheal Intubation*.
    ii. Emergency cricothyroidotomy can be considered to secure airway if severe upper airway edema prevents access to glottic opening with the use of a video laryngoscope per *GUID-320-PR007 – Cricothyrotomy – Surgical*.

f. If the patient is experiencing dyspnea (with or without wheezing), administer
    i. High flow oxygen
    ii. Inhaled bronchodilators and repeat as needed: **Albuterol 2.5 mg/3 ml NSS** and **Ipratropium Bromide 0.02% 0.5 mg/2.5 ml via NEB.**
        1. May repeat as needed **per** *GUID3203-PR012 - Inhalation Therapy*.
    iii. Do not delay intubation if there is rapidly progressing airway edema

g. Consider antihistamine **Diphenhydramine 1 mg/kg IVP/IO/IM.** Max dose 50 mg. DO NOT administer to premature infants or newborns.

h. When hemodynamically stable, consider **Methylprednisolone 1 mg/kg IV.** Maximum dose 60mg or **Dexamethasone 0.6 mg/kg IV or IM** (Max dose 10mg)

i. Consider administering H2 blocker if at sending facility
    i. **Famotidine 0.5 mg/kg IV** (Maximum dose 20 mg) over 2 minutes

IV. **DOCUMENTATION:**

a. EMS Charts

V. **REFERENCES:**

a. Cheng, A., Canadian Paediatric Society, Acute Care Committee. (2011). Emergency treatment of anaphylaxis in infants and children. *Paediatr Child Health*, *16*(1), 35-40.

b. Weingart, S. D. & Borshiff, D. C. (2018). *The Resuscitation Crisis Manual*. Leeuwin Press.